

Lecture 12 - Economics and Business Administration
Cash Flow Statement

The Need to Explain a Change in Cash

BALANCE SHEET

ASSETS	Year x	Year x+1	LIABILITIES	Year x	Year x+1
Cash	100	1,050			
Trade receivables (net)	6,400	9,400	Trade payables	-3,100	-5,700
Inventories	1,200	2,800	Tax payables	-200	-300
Current assets	**7,600**	**12,200**	TFR payables	-2,000	-2,800
			Current liabilities	**-5,300**	**-8,800**
Plant	9,400	8,700	Accrued interest payable	0	-50
Machinery	0	8,000	Dividends payable	0	-100
(Acc. depr. – plant)	-8,600	-8,500	Bank loans	-2,800	0
(Acc. depr. – machinery)	0	-800	Bonds	0	-4,500
Receivables from disposals	0	450	**Financial liabilities**	**-2,800**	**-4,650**
(Payables to plant suppliers)	0	-6,450	Share capital	-300	-450
Fixed assets	**800**	**1,400**	Reserves	-400	-400
Speculative securities	0	300	Net income	-100	-1,000
Equity investments	400	350	**Equity**	**-800**	**-1,850**
Ancillary assets	**400**	**650**			
Total assets	**8,900**	**15,300**	**Total liabilities & equity**	**-8,900**	**-15,300**



CHANGE IN CASH TO BE EXPLAINED = 1,050 – 100 = 950

The cash flow statement reconstructs the underlying reasons for the change experienced by a particular financial resource between the beginning and end of the accounting period



CASH BALANCE

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CASH AND CASH EQUIVALENTS

The forces that influenced any possible change are called FLOWS:

- **flows that increase the fund balance are called SOURCES** 

- **flows that decrease the fund balance are called USES** 

Building the Cash Flow Statement: Overview

- Financial statements do not explicitly show the **cash movements** during the reporting period
- The cash flow statement is built starting from **net income** and working backwards
- Starting from net income, to arrive at the **cash flow** of the period:
  - add back items that **did not generate cash movements**;
  - deduct (add) positive (negative) changes in **operating net working capital**;
  - deduct **cash flows** incurred for **investing activities** (and add those from divestment activities);
  - add/deduct **changes in equity** and financing activities.

Cash Flow Statement Models

- **OBJECT OF ANALYSIS** (fund balance analyzed)
- **STRUCTURE** (logic for aggregating individual flows)
- **PRESENTATION FORMAT**



Object of analysis: *cash flow* (or *cash flow*).

Structure: *4 types of flows*

- **Operating activities cash flow**
- **Investing/divestment activities cash flow**
- **Third-party financing activities cash flow**
- **Equity financing activities cash flow**

Presentation format: *waterfall (indirect method)*

Cash Flows by Activity Type

Operating activities cash flow: flow of financial resources generated or absorbed by **operations attributable to current operating activities** (e.g. collection of sales revenues, payment of operating costs and income taxes)

Investing/divestment activities cash flow: flow of financial resources **absorbed/generated by investing and divestment transactions** (e.g. acquisitions or disposals of tangible fixed assets or non-core assets)

Financing activities cash flow (from third parties and shareholders): flow of financial resources from

a) **drawdowns or repayments of financial debt** (e.g. bank loans or bonds)
b) **equity contributions or distributions to shareholders via dividends**

Construction Methodology

The cash flow statement can be prepared using two different approaches:

- **detailed reconstruction of individual elementary flows**
  DIRECT METHOD

- **determination of aggregate flows by indirect means**
  INDIRECT METHOD

In both cases, the required documents for its preparation consist of the **financial statements**, complete with **notes to the financial statements** and the balance sheet values at the beginning of the period

It is also appropriate that the financial statements schedules be reclassified using the criterion of **CAPITAL EMPLOYED APPROACH**

Exercise

BALANCE SHEET

ASSETS	Year x	Year x+1	LIABILITIES	Year x	Year x+1
CASH	**100**	**1,050**			
			Trade payables	-3,100	-5,700
Trade receivables (net)	6,400	9,400	Tax payables	-200	-300
Inventories	1,200	2,800	TFR payables	-2,000	-2,800
CURRENT ASSETS	**7,600**	**12,200**	**CURRENT LIABILITIES**	**-5,300**	**-8,800**
Plant	9,400	8,700	Accrued interest payable	0	-50
Machinery	0	8,000	Dividends payable	0	-100
(Acc. depr. – plant)	-8,600	-8,500	Bank loans	-2,800	0
(Acc. depr. – machinery)	0	-800	Bonds	0	-4,500
Receivables from disposals	0	450	**FINANCIAL LIABILITIES**	**-2,800**	**-4,650**
(Payables to plant suppliers)	0	-6,450			
FIXED ASSETS	**800**	**1,400**	Share capital	-300	-450
Speculative securities	0	300	Reserves	-400	-400
Equity investments	400	350	Net income	-100	-1,000
ANCILLARY ASSETS	**400**	**650**	**EQUITY**	**-800**	**-1,850**
TOTAL ASSETS	**8,900**	**15,300**	**TOTAL LIABILITIES & EQUITY**	**-8,900**	**-15,300**

CHANGE IN CASH TO BE EXPLAINED = 1,050 – 100 = 950

Exercise

INCOME STATEMENT – Year X+1

Opening inventories	1,200	Revenues	-22,600
Purchases and other op. costs	16,000	Closing inventories	-2,800
Lease costs	1,400	Gain on disposal	-1,100
Labour costs	4,200	Interest income on securities	−15
TFR accrual	900		
Depreciation − plant	500	**Total positive components**	**−26,515**
Depreciation − machinery	800		
Write-down of investments	50		
Interest expenses	165		
Income taxes	300		
Total negative components	***25,515***		
NET INCOME	**1,000**		
TOTAL	*26,515*	*TOTAL*	*26,515*

Exercise: Information from the Notes to the Financial Statements

1) An investment was written down by 50

2) A plant was sold:
- - historical cost: 700
- - accumulated depreciation: 600
- - sale price: 1,200; at year-end a receivable of 450 remains outstanding.

3) A new machinery has been purchased for 8.000; at the end of the year there is an account payable of 6.450, to be paid during year X+2.

4) Excess cash was invested in government securities (classified under non-core assets)

5) The bank loan was repaid for 2,800.

6) A bond was issued for 4,500.

7) Dividends of 100 were declared.

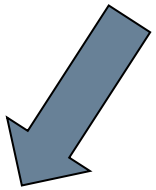
8) A share capital increase at par value was carried out; shareholders paid in 150.

9) Severance pay (TFR) was paid to resigning employees

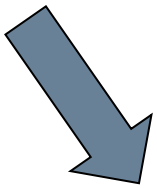
Building the Cash Flow Statement: Indirect Method

Identification of the cash change to be explained:

+ Closing cash :	**1,050**
- Opening cash :	**100**
= Δ LIQUIDITY' :	**950**



Cash change attributable to **operating activities**



Cash change attributable to:

- **investing/divestment activities**
- **financing activities**

Operating Activities Cash Flow

Δ LIQUIDITY ATTRIBUTABLE TO CURRENT OPERATING ASSETS

Assumption: net income for the period translates entirely into cash flows

1. Correcting the assumption by adjusting net income for items that **have not generated and will not generate any cash inflow/outflow** (**non-cash income and expenses**);

2. Net income is adjusted for items that **do not relate to current operating activities, but to investing/divestment or financing activities** (**income and expenses belonging to other activity areas**)

Adjusted Net Income*

3. Correcting the initial assumption by accounting for **revenues not yet collected and costs not yet paid** (**changes in operating NWC**)

** Adjusted Net Income: net income restated to remove non-cash items and income/expenses not related to current operating activities, representing the cash the company is able to spend from operations.*

Operating Activities Cash Flow (1)

1. Elimination of “non-cash cost and revenues”.
   Consider the following equation:

Positive income components	-	Negative income components	=	Net income

which can be rewritten as

Cash revenues	+	Non-cash revenues	-	Cash costs	-	Non-cash costs	=	Net income

from which it follows…..

Cash revenues	-	Cash costs	=	Net income	+	Non-cash costs	-	Non-cash revenues

E.g. Depreciation, amortization and write-downs of fixed assets

E.g. Revaluation of fixed assets

Operating Activities Cash Flow (2)

+/- Net income	1,000	
+ Depreciation & amortization	1,300	non-cash income and expenses
+ Write-down	50	
- Gain on disposal	-1,100	income and expenses not related to current operating activities
+ Interest expenses	165	
= ADJUSTED NET INCOME*	1,415	

This adjustment is not a reversal, as in the previous case, but a reclassification!

2. **Elimination of income and expense items that do not belong to current operating activities.**

Adjusted Net Income* is the potential cash generated by core operating activities!

** Adjusted Net Income: net income restated to remove non-cash items and income/expenses not related to current operating activities, representing the cash the company is able to spend from operations.*

Operating Activities Cash Flow (3)

3. Were all revenues collected? Were all costs fully paid?

Change in operating NWC

- 1,100

= ADJUSTED NET INCOME*	1,415	
+ Opening trade receivables	6,400	(3,000)
- Closing trade receivables	(9,400)	
+ Opening inventories	1,200	(1,600)
- Closing inventories	(2,800)	
- Opening trade payables	(3,100)	2,600
+ Closing trade payables	5,700	
- Opening tax payables	(200)	100
+ Closing tax payables	300	
- Opening TFR payables	(2,000)	800
+ Closing TFR payables	2,800	
= CASH GENERATED (ABSORBED) BY CURRENT OPERATING ACTIVITIES	315	

** Adjusted Net Income: net income restated to remove non-cash items and income/expenses not related to current operating activities, representing the cash the company is able to spend from operations.*

The Change in Operating NWC: The "Sponge Effect"



When NWC increases, cash is absorbed





When NWC decreases, cash is released



Investing Activities Cash Flow

The cash flow from **investing and divestment activities** is determined by observing movements in **core fixed operating assets** (tangible and intangible fixed assets) and in **ancillary activities** (financial and real estate assets), as well as related movements in payables to suppliers of fixed assets and receivables from disposals

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disposal of tangible, intangible and financial assets (including related gains and losses, already reversed from operating income), net of any changes in receivables from such disposals

NEUTRAL TRANSACTIONS

contribution of assets against equity, assets received as donations, write-downs

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purchase of tangible, intangible and financial assets, net of any changes in payables from such purchases

Investing Activities Cash Flow

- Non Core activity investments	- 300
- core activity investments	- 8,000
- opening payables to plant suppliers	0
+ closing payables to plant suppliers	+ 6,450
= total investments	- 1,850
- core activity disposals	+ 1,200
+ opening receivables from plant disposals	0
- closing receivables from plant disposals	- 450
= total disposals	+ 750

1,100 gain on disposal

100 net book value

- An increased in a fixed asset represents a use of financial resources; payables to suppliers, by contrast, represent a source;
- A decrease in a fixed asset, net of any related receivables, is a source of cash.

Financing Activities Cash Flow

Cash changes from financing activities are attributable to transactions that change financial liabilities and/or equity



drawdown of loans, increase of other financing, paid-in capital increases



repayment of financial liabilities, payment of dividends or interest (net of accruals and payables), return of equity capital

- Repayment of financial loans	-2,800
+ Drawdown of financial loans	4,500
- Interest expenses	-165
+/- Change in accrued interest payable	50
= Changes in financial debt	**1,585**
+ Capital increase	150
- Dividend distribution	0
= Changes in equity	**150**
= CASH FROM FINANCING ACTIVITIES	**1,735**

Building the Cash Flow Statement: Indirect Method

Cash – Year X		100
Cash – Year X+1		1,050
CASH FLOW TO BE EXPLAINED		**950**
+ Net income	1,000	
+ Depreciation & amortization	1,300	
+ Write-down of investments	50	
- Gain on disposal	-1,100	
+ Interest expenses	165	
= Distributable income	***1,415***	
+/- Change in trade receivables (net)	-3,000	
+/- Change in inventories	-1,600	
+/- Change in trade payables	2,600	
+/- Change in tax payables	100	
+/- Change in TFR payables	800	
= Change in operating NWC	***-1,100***	
= CASH FROM OPERATING ACTIVITIES		**315**
- Ancillary activity investments	-300	
- Core activity investments	-8,000	
+/- Change in payables to plant suppliers	6,450	
= Investments	***-1,850***	
+ Disposals	1,200	
+/- Change in receivables from disposals	-450	
= Divestments	***750***	
= CASH USED IN INVESTING ACTIVITIES (CAPEX)		**-1,100**
- Repayment of financial loans	-2,800	
+ Drawdown of financial loans	4,500	
- Interest expenses	-165	
+/- Change in accrued interest payable	50	
= Changes in financial debt	***1,585***	
+ Capital increase	150	
- Dividend distribution	0	
= Changes in equity	***150***	
= CASH FROM FINANCING ACTIVITIES		**1,735**
= NET CASH GENERATED (ABSORBED) IN THE PERIOD		**950**

Interpreting the Cash Flow Statement

Cash flow analysis allows observation of financial equilibrium conditions, i.e. the **company's ability to generate sufficient cash flows to cover its needs** (balance between sources and uses)

A company achieves **financial equilibrium** not only by having a **positive overall cash flow** (sources exceeding uses) but also when there is **balance among different types of flows**

It is useful to distinguish between:

- **RECURRING** sources and uses of cash, expected to occur in every period, even if with different amounts;
- **NON-RECURRING** or sometimes exceptional, which will not necessarily recur in every period, as they result from discretionary decisions that the company may take only occasionally.

Interpreting the Cash Flow Statement

Once prepared, the cash flow statement must be analysed and its findings related to those from income and balance sheet analysis. For example:

The overall change in cash is positive because:

- There are significant sources from **disposal of investments generating significant gains** …
- There is a **decrease in Operating NWC** driven by a sharp reduction in inventories and a significant increase in trade payables…
- To fund investments, **short-term loans have been raised** …
- There has been a significant **capital increase**…

The overall change in cash is negative because:

- The company is **growing** (in revenue terms) and operating NWC is absorbing cash…
- The company is investing in **new production plants** and retail outlets…
- The company is **reducing its financial debt**…

Focus on Operating Activities

Consider two companies in the same industry with the same **level of revenues and fixed asset investments** (i.e. same operational and structural scale): looking at their operating cash flow, which appears to be performing better?

	Company A	Company B
Adjusted Net income	**+ 100**	**+ 300**
Change in operating NWC	**0**	**-200**
Operating activities cash flow	**= 100**	**= 100**

Focus on Operating Activities

Consider two companies in the same industry with the same **level of revenues and fixed asset investments** (i.e. same operational and structural scale): looking at their operating cash flow, which appears to be performing better?

	Company A	Company B
Adjusted Net income	**+ 100**	**+ 300**
Change in operating NWC	**0**	**-200**
Operating activities cash flow	**= 100**	**= 100**

Cash Flow Analysis: Ratios and Metrics

Cash flow analysis aims to understand not only the balance among different flow types, but also the **magnitude of those flows**. This analysis is often carried out using specific ratios that:

- compare cash flows of different types;
- compare cash flows with income statement or balance sheet figures

$$\textit{Free Cash Flow} = \frac{\textbf{Cash flow from operating activities}}{\textbf{Capital expenditures on operating fixed assets, net of disposals (CAPEX).}} = \frac{315}{1.100} = 28.64\%$$

Cash Flow Analysis: Ratios and Metrics

Free Cash Flow

The company's ability to **recover its investments through operating cash flows**

Debt Repayment Capacity

$$\frac{\text{Total financial liabilities}}{\text{Operating activities cash flow}} \quad \frac{4{,}650}{315} = 14.8$$

No. of years the operating cash flow will take to repay outstanding debt ≈ NFD / EBITDA

Cash Flow Analysis: Ratios and Metrics

Cash Profitability

$$\frac{\text{Operating activities cash flow}}{\text{Net sales}} \qquad \frac{315}{22{,}600} = 1.39\%$$

For every €100 of revenue, the % of cash generated.
The financial equivalent of ROS.

Earnings Quality Ratio

$$\frac{\text{Operating activities cash flow}}{\text{Net income}} \qquad \frac{315}{1{,}000} = 31.5\%$$

Measures the gap between financial and accounting performance

Concluding Remarks

The financial position is not as healthy as it appears - the balance sheet reclassified by current/non current approach shows this

BALANCE SHEET

ASSETS	Year x	Year x+1	LIABILITIES	Year x	Year x+1
Cash	100	1,050	Trade payables	-3,100	-5,700
Speculative securities	0	300	Payables to plant suppliers	0	-6,450
Receivables from disposals	0	450	Tax payables	-200	-300
Trade receivables (net)	6,400	9,400	Loans due within 12 months	-2,800	-1,000
Inventories	1,200	2,800	Accrued interest payable	0	-50
			Dividends payable	0	-100
CURRENT ASSETS	**7,700**	**14,000**	**CURRENT LIABILITIES**	**-6,100**	**-13,600**
Plant	9,400	8,700	TFR payables	-2,000	-2,800
Machinery	0	8,000	Bonds	0	-3,500
(Acc. depr. - plant)	-8,600	-8,500	**NON-CURRENT LIABILITIES**	**-2,000**	**-6,300**
(Acc. depr. - machinery)	0	-800	Share capital	-300	-450
Equity investments	400	350	Reserves	-400	-400
NON-CURRENT ASSETS	**1,200**	**7,750**	Net income	-100	-1,000
			EQUITY	**-800**	**-1,850**
TOTAL ASSETS	**8,900**	**21,750**	**TOTAL LIABILITIES & EQUITY**	**-8,900**	**-21,750**

- **Current Assets – Current Liabilities 1,600 (X) → 400 (X+1)**
- **Net Working Capital is significantly deteriorating!**